

SOLUTIONS

Full marks are not necessarily awarded for a correct answer with no working. Answers must be supported by working and/or explanations. Where an answer is incorrect, some marks may be given for a correct method, provided this is shown by written working. You are therefore advised to show all working.

Section A

Answer **all** questions. Answers must be written within the answer boxes provided. Working may be continued below the lines, if necessary.

1. [Maximum mark: 5]

Point P has coordinates $(-3, 2)$, and point Q has coordinates $(15, -8)$. Point M is the midpoint of [PQ].

(a) Find the coordinates of M. [2]

Line L is perpendicular to [PQ] and passes through M.

(b) Find the gradient of L. [2]

(c) Hence, write down the equation of L. [1]

$$(a) \quad M = \left(\frac{-3+15}{2}, \frac{2+(-8)}{2} \right) \\ = (6, -3) \quad \text{A1 A1}$$

$$(b) \quad m_{PQ} = \frac{-8-2}{15-(-3)} = \frac{-10}{18} = -\frac{5}{9} \quad \text{(A1)} \\ m_L = \frac{9}{5} \quad \text{A1 A1}$$

$$(c) \quad y - (-3) = \frac{9}{5}(x - 6) \\ y + 3 = \frac{9}{5}(x - 6) \quad \text{A1}$$

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2. [Maximum mark: 7]

The function f is defined by $f(x) = \frac{7x+7}{2x-4}$ for $x \in \mathbb{R}$, $x \neq 2$.

- (a) Find the zero of $f(x)$. [2]
- (b) For the graph of $y = f(x)$, write down the equation of
- (i) the vertical asymptote;
 - (ii) the horizontal asymptote. [2]
- (c) Find $f^{-1}(x)$, the inverse function of $f(x)$. [3]

(a) $f(x) = \frac{7x+7}{2x-4} = 0$ (m1)
 $7x+7 = 0$
 $x = -1$ A1

(b) i) $x = 2$ A1
 ii) $y = 7/2$ A1 (equations)

(c) $f^{-1}: x = \frac{7y+7}{2y-4}$ (m1)
 $2xy - 4x = 7y + 7$
 $2xy - 7y = 4x + 7$ (A1)
 $f^{-1}: y = \frac{4x+7}{2x-7}$ $x \neq \frac{7}{2}$ A1

⑥



3. [Maximum mark: 6]

On a Monday at an amusement park, a sample of 40 visitors was randomly selected as they were leaving the park. They were asked how many times that day they had been on a ride called *The Dragon*. This information is summarized in the following frequency table.

Number of times on <i>The Dragon</i>	Frequency
0	6
1	16
2	13
3	2
4	3

It can be assumed that this sample is representative of all visitors to the park for the following day.

(a) For the following day, Tuesday, estimate

(i) the probability that a randomly selected visitor will ride *The Dragon*;

(ii) the expected number of times a visitor will ride *The Dragon*.

[4]

It is known that 1000 visitors will attend the amusement park on Tuesday. *The Dragon* can carry a maximum of 10 people each time it runs.

(b) Estimate the minimum number of times *The Dragon* must run to satisfy demand.

[2]

(a) i) $P(\text{Dragon}) = \frac{40-6}{40} = \frac{34}{40}$ (M1) A1

ii) $E(X) = (6 \cdot 0 + 16 \cdot 1 + 13 \cdot 2 + 2 \cdot 3 + 3 \cdot 4) / 40$ (M1)

$= (16 + 26 + 6 + 12) / 40$

$= 60 / 40$

$= 1.5$ A1

(b) # rides = $1000 \cdot 1.5$ (M1)

$= 1500$

runs = $\frac{1500}{10} = 150$ A1

6



4. [Maximum mark: 6]

(a) Show that the equation $\cos 2x = \sin x$ can be written in the form $2 \sin^2 x + \sin x - 1 = 0$. [1](b) Hence, solve $\cos 2x = \sin x$, where $-\pi \leq x \leq \pi$. [5]

$$(a) \dots \cos 2x = 1 - 2 \sin^2 x \dots (\text{identity})$$

$$\dots 1 - 2 \sin^2 x = \sin x \dots \text{A1}$$

$$\dots 2 \sin^2 x + \sin x - 1 = 0$$

$$(b) \dots (2 \sin x - 1)(\sin x + 1) = 0 \dots (m1)$$

$$\dots 2 \sin x - 1 = 0 \quad \text{or} \quad \sin x + 1 = 0$$

$$\dots \sin x = \frac{1}{2} \quad \sin x = -1 \dots (m1)$$

$$\dots x = \frac{1}{6}\pi, \frac{5}{6}\pi \quad x = -\frac{\pi}{2} \quad \text{A1}$$

$$x = -\frac{\pi}{2}, \frac{1}{6}\pi, \frac{5}{6}\pi \quad \text{A1}$$



6



(Question 1 continued)

$$(a) \quad P = 4 + 4 + 10 \\ = 18 \text{ cm}$$

(M1)
A1

$$(b) \quad \theta = \frac{10}{4} = 5/2$$

(A1) A1

$$(c) \quad A = \left(\frac{1}{2}\right) \frac{5}{2} (4^2)$$

(M)

$$= \frac{5}{18} \times 20 \text{ cm}^2$$

A1



16EP03

Turn over

3. [Maximum mark: 5]

A function f is defined by $f(x) = 1 - \frac{1}{x-2}$, where $x \in \mathbb{R}$, $x \neq 2$.

- (a) The graph of $y = f(x)$ has a vertical asymptote and a horizontal asymptote.

Write down the equation of

- (i) the vertical asymptote;
(ii) the horizontal asymptote.

[2]

- (b) Find the coordinates of the point where the graph of $y = f(x)$ intersects

- (i) the y -axis;
(ii) the x -axis.

[2]

(a) i) $x = 2$
ii) $y = 1$

A1
A1

(b) i) $f(0) = 1 - \frac{1}{0-2} = \frac{3}{2}$
 $(0, \frac{3}{2})$

A1

ii) $f(x) = 1 - \frac{1}{x-2} = 0$
 $x-2 = 1$
 $x = 3$
 $(3, 0)$

A1

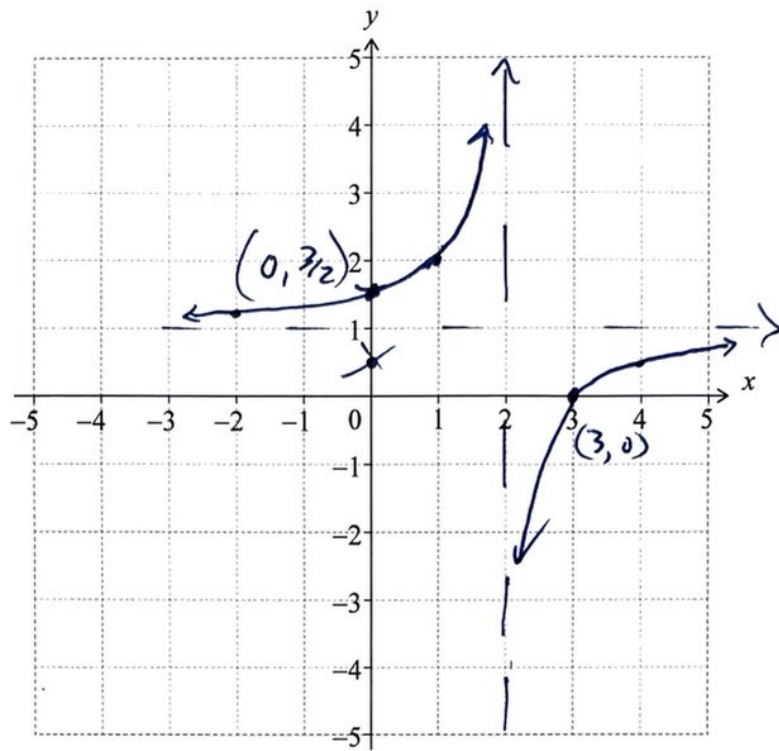
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(Question 3 continued)

- (c) On the following set of axes, sketch the graph of $y = f(x)$, showing all the features found in parts (a) and (b).

[1]



5. [Maximum mark: 6]

Find the range of possible values of k such that $e^{2x} + \ln k = 3e^x$ has at least one real solution.

$$e^{2x} - 3e^x + \ln k = 0$$

$$u = e^x \quad (M1)$$

$$e^x = \frac{3 \pm \sqrt{(-3)^2 - 4(1)(\ln k)}}{2}$$

$$\begin{aligned} a &= 1 \\ b &= -3 \\ c &= \ln k \end{aligned}$$

$$\frac{3 \pm \sqrt{9 - 4\ln k}}{2} > 0$$

$$\begin{aligned} e^x &> 0 \quad \text{AND} \\ \Delta &\geq 0 \quad (M1) \end{aligned}$$

$$3 > \sqrt{9 - 4\ln k}$$

$$9 > 9 - 4\ln k$$

$$\begin{aligned} 4\ln k &> 0 \\ k &> 0 \end{aligned}$$

$$\Delta = 9 - 4\ln k \geq 0 \quad (A1)$$

$$\frac{9}{4} \geq \ln k \quad (A1)$$

$$k \leq e^{\frac{9}{4}} \quad A1$$

$$0 < k \leq e^{\frac{9}{4}} \quad A1$$

6



7. (a) $h(0) = 2(0)e^0 + 3$ (M1)
 $y = 3$ A1

(b) $h'(x) = 2xe^x + 2e^x$ (M1) A1

(c) $2xe^x + 2e^x = 0$ (M1) (A1)
 $x = -1$ A1
 $h(-1) = 2(-1)e^{-1} + 3$ (M1)
 $= -2e^{-1} + 3$
 $(-1, -2e^{-1} + 3)$ A1

(d) $h'(x) = 2xe^x + 2e^x$

(i) $h''(x) = (2xe^x + 2e^x) + 2e^x$ A1 A1
 $= (2x + 4)e^x$

(ii) $h''(x) = (2x + 4)e^x > 0$ (M1)
 $2x + 4 > 0$
 $x > -2$ A1

P1 T21

SOLUTIONS

8(a)

$$(i) S_5 = 5^2 + 4(5) \\ = 45$$

(m1)
A1

$$(ii) u_6 = S_6 - S_5 \\ = 60 - 45 \\ = 15$$

(m1)

A1

$$(b) S_6 = \frac{6}{2}(u_1 + 15) = 60 \\ u_1 = 5$$

(m1)
A1

$$(c) u_6 = 5 + d(6-1) = 15 \\ d = 2$$

(m1)

(A1)

$$u_n = 5 + 2(n-1)$$

A1

$$(d) v_2 = 5, v_4 = 15 \\ r^2 = \frac{15}{5} = 3 \\ r = \pm \sqrt{3}$$

(m1)(A1)
A1E

$$(e) v_{99} = 5 \cdot r^{(99-2)} < 0 \\ r^{97} < 0 \\ r = -\sqrt{3}$$

(m1)

$$v_5 = 15 \cdot (-\sqrt{3}) \\ = -15\sqrt{3}$$

A1

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7. (a) $x = -2$

A1

(b) $h = -2$

A1 A1

$k = -5$

(c) $f(0) = \frac{1}{4}(0+2)^2 - 5$
 $y = -4$

(M1)
 A1

(d) $f'(x) = \frac{1}{4}(2)(x+2)$
 $= \frac{1}{2}x + 1$

(A1)

$f'(0) = \frac{1}{2}(0) + 1 = 1$

(M1)

$m_{\perp} = -1$

A1

$y - (-4) = -1(x - 0)$

$y = \cancel{-x-4} -x-4$

A1

(e) Q: $\frac{1}{4}(x+2)^2 - 5 = -x - 4$

(M1)

$x^2 + 4x + 4 = -4x + 4$

$x^2 + 8x = 0$

(A1)

$x = 0, -8$

A1

$y = -(-8) - 4 = 4$

(M1)

Q $(-8, 4)$

A1

$d_{PQ} = \sqrt{(0 - (-8))^2 + (-4 - 4)^2}$
 $= 8\sqrt{2}$

(A1)

A1

9. (a) Q: $x^2 + y^2 = 9$

$$y^2 = 9 - x^2$$

$$y = \sqrt{9 - x^2}$$

$y > 0$ given 1st Quadrant

(b) $A_0 = \frac{1}{2}(2)(-3-x)\sqrt{9-x^2}$
 $= (x+3)\sqrt{9-x^2}$

(A1)(M1)
A1

(c) $\frac{dA}{dx} = (x+3)\left(\frac{1}{2}\right)\left(\frac{1}{2}(9-x^2)^{-\frac{1}{2}}\right)(-2x) + (1)\sqrt{9-x^2}$

(M1)(M1)

$$= \frac{-x(x+3)}{\sqrt{9-x^2}} + \frac{(\sqrt{9-x^2})^2}{\sqrt{9-x^2}}$$

A1

$$= \frac{-x^2 - 3x + 9 - x^2}{\sqrt{9-x^2}}$$

A1

$$= \frac{9 - 3x - 2x^2}{\sqrt{9-x^2}}$$

(4)

(d) $A_{max}: \frac{dA}{dx} = \frac{9 - 3x - 2x^2}{\sqrt{9-x^2}} = 0$

(M1)

$$(3+x)(3-2x) = 0$$

(A1)

(M1)
disregard $x = -3$

$$y = \pm \sqrt{9 - \left(\frac{3}{2}\right)^2} \quad x = \frac{3}{2}$$

A1

(M1)

R_y:

$$= -\frac{3}{2}\sqrt{3}$$

A1